

Verification of music21 Chord Symbol Parsing (Latest Version vs. Guide)

Parsing Rules and Format in music21

Music21's `harmony.ChordSymbol` parser uses **the exact order and syntax** described in the “コードシンボル完全ガイド.” The chord name must follow a fixed sequence with **no extra characters** (no spaces, commas, or parentheses) ¹. The required format is:

- **Root and Accidental:** A letter A–G, optionally followed by an accidental. Use `#` for sharp and `-` (hyphen) for flat **in the root or bass only** ². For example, B ♭ major is written as `B-` (not “Bb”).
- **Quality (Chord Type):** Optional chord quality/kind symbols immediately after the root. The parser recognizes standard abbreviations: e.g. `maj` or `M` for major, `m` or `min` for minor, `dim` / `o` for diminished, `aug` / `+` for augmented, `sus4` / `sus2` for suspensions, `mM` for minor-major, `+M7` for augmented-major seventh, `power` for fifth chords, etc ³. If no quality is given, a major triad is assumed ⁴.
- **Extension (7, 9, 11, 13):** Optional numeral indicating seventh or extended chord. If a 9th, 11th, or 13th is specified, music21 **automatically includes the 7th** in the chord structure by rule ⁵. (E.g. `C9` is understood as C7 with an added 9th.) For **Major 9th/11th/13th chords**, you must include a `maj` quality so the parser knows it's a major 7th chord ⁵. (For instance, `Cmaj9` means C–E–G–B–D, whereas `C9` would be treated as a dominant 9th C–E–G–B ♭–D.)
- **Alterations (`#` / `b` tensions):** Altered tensions are added after the extension. Use `#` or `b` followed by the scale degree number, with no separators ⁶. For example, `C7b9#11` is valid (meaning a C7 with a flat 9th and sharp 11th) ⁷. **Do not use commas or parentheses** between multiple alterations – just string them together (the parser will accept multiple alterations in any order) ⁶. Important: In this context, use `b` for flat (e.g. `b9`), **not** a hyphen. Writing `C7-9` will cause an error – it must be `C7b9` ⁸. Likewise `Gm7-5` must be `Gm7b5`, etc. ⁸.
- **Additions and Omissions:** You can specify added or omitted tones using `addN` or `omitN`. These should come after any alterations. For example, `Cadd9` (add a 9th to a triad) or `C7omit3` (omit the 3rd from a C7 chord) are legal ⁹. Again, **no parentheses** – e.g. write `Amadd9`, not “Am(add9)” ¹⁰.
- **Suspensions (sus):** If the chord quality already includes a third, you can indicate a suspension by using a `sus` chord type (e.g. `Csus4`, `D7sus4`). The parser recognizes `sus4` / `sus2` as chord qualities (and even `7sus4` as a built-in seventh chord type) ¹¹. Additionally, the guide notes you may reapply a `sus` after extensions – for example, writing `C9sus4` to indicate a 9th chord with no third and a suspended 4th ¹². In practice, music21's dictionary doesn't directly have a “9sus4” quality, so to get a ninth with a suspension you should combine forms (e.g. `C7sus4add9` which the parser will understand as C7sus4 with an added 9th). The key point is that **“sus” must appear as part of the chord symbol string** (not in parentheses) to be parsed.
- **Bass Slash:** An alternate bass is indicated with a slash followed by a pitch. This comes at the very end of the chord symbol. The bass letter must also be A–G with optional `#` or `-`. For instance, `Fsus2/C` or `G7/D-` (G7 over D ♭ bass) are valid slash chords ¹³.

Summary: The latest music21 version (2025) still adheres to this strict regex-style format – **Root → Accidental → Quality → Extension → Alterations → add/omit → sus → /Bass** – exactly as outlined in the guide ¹⁴ ¹⁵. Any deviation (extra characters, out-of-order terms, or unrecognized abbreviations)

will lead to a parsing failure. The guide's warnings about forbidden characters and fixed ordering remain 100% applicable in the current music21 parser.

Ambiguous or Error-Prone Chord Examples and Corrections

Cmaj13#11 – Major 13th chord with a sharp 11th:

- **Accepted? Yes**, this can be interpreted by music21, but you must not omit the “maj” and need to be mindful of spelling. `Cmaj13#11` is meant to denote a C major - 13 chord (which includes a major 7th) with an altered 11th. In the latest version, it will parse provided the quality is recognized. The parser sees “maj13” and knows it's a major 13th chord (C–E–G–B–D–F–A with F raised to F# for the #11) ¹⁶. However, in some earlier versions there was a quirk: the abbreviation `maj13` in lowercase wasn't in the default dictionary (it expected `Maj13` with a capital “M”) ¹⁷. If you encounter a `ValueError`, it's likely due to that casing issue.

- **Correct Notation:** `Cmaj13#11` (with “maj”) is the intended correct format. If for some reason that isn't accepted, an alternative is to explicitly spell it as `Cmaj7#11add13`. This format makes the major 7th explicit and then adds the 13th, which the parser will definitely understand ¹⁷. (For example, the guide suggests that `Fmaj13` was not recognized and recommends `Fmaj7add13` instead ¹⁷.)

- **Why Problems Occur:** The parser needs to know it's a **major-type chord** when you use 9, 11, or 13. Forgetting “maj” (e.g. writing `C13#11`) would be read as a dominant 13th (with a flatted 7th) – a different chord. Also, writing “maj13” in an unrecognized way (like an unexpected case) can confuse the lookup. In short, **the chord must explicitly indicate the major quality** for extended chords, and the syntax must match one of the parser's known patterns. When properly notated, `Cmaj13#11` is parsed as a valid chord in the latest music21.

G7alt – G dominant with “alt” alterations:

- **Accepted? No – not as-is.** The string `G7alt` will **not** be parsed by music21's `ChordSymbol`. The term “alt” (short for “altered”) is not in the library's default chord vocabulary ¹⁸. If you try `harmony.ChordSymbol("G7alt")`, it will throw an error saying the chord type cannot be identified.

- **Correct Notation:** You must spell out the specific alterations. In jazz, “7alt” implies a dominant seventh with altered 5th and 9th (usually interpreted as having $\flat 5 / \sharp 5$ and $\flat 9 / \sharp 9$, though definitions vary). To be explicit, you could write `G7  $\flat 5$   $\flat 9$` , `G7  $\sharp 5$   $\sharp 9$` , or whatever set of altered tensions you intend. For a fully altered chord (all 5th and 9th variations), one valid spelling is `G7 $\flat 9$  $\sharp 9$  $\sharp 11$  $\flat 13$` ¹⁹. For example, `G7 $\sharp 5$  $\sharp 9$` is a common choice if you mean a dominant with sharpened 5th and 9th. Any of these explicit forms will be accepted by the parser (since it sees known alterations like $\flat 9$, $\sharp 5$, etc.). You can then **label** the chord “G7alt” in output if needed, but the input to music21 must be the detailed form.

- **Why it Fails:** The parser relies on a dictionary of chord types and abbreviations. “Alt” was never added to that dictionary ¹⁸. The guide confirms that the “alt” notation isn't recognized by music21 and should be replaced by concrete alterations ¹⁸. In short, music21 doesn't attempt to guess what “alt” means – you have to tell it each altered scale degree. (Similarly, the triangle symbol $\triangle$ for major is not recognized either, for the same reason of not being in the default chord symbol list ¹⁸.)

Am(add9) – A minor with added 9th in parentheses:

- **Accepted? No.** The format `Am(add9)` will cause a parse error. Parentheses **are not allowed** in chord symbols (and are unnecessary) ¹⁰. Music21 will stop at the unexpected “(” character and fail to interpret this chord name.

- **Correct Notation:** Write `Amadd9` instead. This is the proper way to indicate an added 9th on an A minor triad. For example, `Amadd9` means the pitches A–C–E (minor triad) plus a B (the 9th). If you actually intend a minor 9th chord (which includes a flat 7th, G), then use `Am9` – but note that “Am9” assumes a G is present as it's a minor 7th chord extended to the 9th. For just adding the 9 to a triad (no 7th), `Amadd9` is the clearest.

- **Why it Fails:** The music21 parser expects the chord symbol string to be continuous with no extraneous characters. Parentheses and commas are explicitly disallowed ¹ ¹⁰. In the guide's list of common errors, `Am(add9)` is cited as a notation that will throw a `ValueError`, and the solution is to remove the parentheses and write `Amadd9` (or `Am9`) directly ¹⁰. Essentially, the parser doesn't treat "(" as meaning anything – it just breaks the pattern – so the entire chord name can't be understood until it's formatted without those characters.

Updates in Recent Versions and Additional Notes

Overall, the latest music21 version (2025) has the same fundamental parsing rules as described in the guide. The strict syntax (no spaces or special characters) and the required order of components have not changed. A few points to note:

- **No New Shorthands:** Common shorthand symbols that were unsupported in earlier versions remain unsupported unless you manually extend the dictionary. For example, the “**alt**” shorthand is still not recognized by default (you must spell out alterations) ¹⁸. The triangle **Δ** for major seventh is also not recognized – use “maj7” or “M7” instead ¹⁸. There haven't been new additions like these to the chord syntax as of 2025, so the guide's caution to avoid them is still valid. (If you absolutely need them, you can use `harmony.ChordSymbol.addNewChordSymbol()` to define custom aliases, but that is an advanced workaround outside the normal parser usage.)
- **Accidental Handling:** The flat (`-`) vs. sharp (`#`) rules are unchanged. The root or bass note **must** use `-` for flat, while **alterations use** `b`. This dichotomy remains a common source of user error. The parser will throw an error if you use a wrong symbol (e.g., “E7-9” with a hyphen flat in the extension) ⁸. Always write `b5`, `b9`, etc., and use hyphen only immediately after the root letter or after a slash root in inversions ².
- **Case Sensitivity and Implied Defaults:** The chord abbreviations are generally case-insensitive for the root note, but the chord quality part often **must match one of the known patterns exactly**. The guide pointed out that “maj” with extensions could be tricky – e.g., “maj7” is recognized, and “Maj9” is recognized, but a lowercase “maj9” might not have been in the list ¹⁷. Recent versions have largely standardized this, but if you hit an “Invalid chord abbreviation” error, check if maybe the capitalization is off. In practice, writing `maj` in lowercase for 7th chords is fine (`Cmaj7` works), and for 9th/13th chords music21 now accepts both forms (you can use `Cmaj9` and `Cmaj13` and it should parse in v7+). Just be aware that internally it's matching against a list of abbreviations (like “Maj13” or “M13”), so consistency matters.
- **Major 9th/13th Clarification:** One change from older conventions is that music21 requires the “maj” qualifier on extended major chords. In the past, some people might write “Fmaj9” or “Fmaj13” interchangeably with “FΔ9”. Music21 will parse `Fmaj9` (meaning Fmaj7add9) correctly in the latest version, but you **cannot** write `F9` if you mean a major 9th – `F9` will be taken as dominant 9th (flat 7th). The guide emphasizes using `maj7add9` if needed ¹⁷, but as of now `Fmaj9` is supported as a single token chord name (yielding F–A–C–E–G). The key point: always include the “maj” for chords that have a major seventh with extensions, because the parser won't infer a major 7th from just “9” or “13” alone.
- **Strict Parsing (Error Prevention):** The latest music21 is slightly **more strict** in enforcing correct syntax – which is a good thing for catching typos. Any leftover or unparsed characters in the chord string will cause a failure. For example, a stray parenthesis or a comma will not be ignored; it will abort the parse. The guide's list of “よくあるエラー” (common errors) and their fixes is still the best reference for these pitfalls. To reiterate a few from the guide that apply exactly the same in 2025: do **not** include commas between alterations (`C7(b9,#11)` is wrong – use `C7b9#11` ¹⁰), do **not** use lowercase “b” for a flat root (`Ebm` is wrong – use `E- m`), do **not** use a minus sign for an altered tension (`-13` is wrong – use `b13` ⁸), and avoid unrecognized combos like `B-maj9#11`

(which needs to be split into either `B-maj7#11add9` or `B-maj13#11` as the guide suggests ¹⁷). The parser will enforce all these rules.

- **Special Cases (No Chord):** Music21 does handle “no chord” indications if you use the proper notation. The guide notes that “**N.C.**” or “**NoChord**” are understood as an indication of no chord/harmonic rest ²⁰. In the library, `harmony.ChordSymbol('N.C.')` produces a special chord symbol (internally it’s treated akin to a rest). This behavior is unchanged – you can use `"N.C."` in a chord list (e.g. in a YAML chord map) to denote a bar of no chord. The parser will not try to find a root and will instead yield a `NoChord` object (or you can insert an actual `note.Rest()` as needed) ²⁰. Just make sure it’s exactly `N.C.` or `NoChord` (case-insensitive) for the rest – do not combine it with other text.

In summary, the **music21 chord symbol parsing in 2025 matches the regex rules given in the complete guide**. The ordering `Root→Quality→Extensions→Alterations→Adds/Omits→Sus→Bass` is strictly enforced ¹⁴. All the cautionary examples in the guide still apply: you must use the correct symbols (hyphens vs. flats, no illegal characters) and the proper names for chord types. The few areas of ambiguity – like the “maj” casing for 13th chords or the unsupported “alt” shorthand – have either been clarified or remain as they were (with “alt” still unsupported by default) ¹⁸. By following the guide’s format to the letter, chord symbols should parse correctly in the latest music21, ensuring your YAML chord maps and other inputs won’t trigger those dreaded `ValueError` exceptions due to formatting issues. **If a chord isn’t parsing, nine times out of ten it’s because it violates one of the above rules.** The guide’s comprehensive list of do’s and don’ts is still the best blueprint for valid chord notation in music21’s current version.

Sources: The music21 コードシンボル完全ガイド provided (which summarizes the regex and rules) ²¹ ²², and music21’s documentation/code confirming unsupported shorthands ¹⁸ and syntax requirements ². The analysis of examples (`Cmaj13#11`, `G7alt`, `Am(add9)`) is based on these rules and the known behavior of `harmony.ChordSymbol` in music21 v7/v8 ¹⁷ ¹⁰.

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<https://github.com/Haluto-oss/my-music-app/blob/38b68f561b9fa0c74f09977f7170c9b3cd733094/backend-python/venv/Lib/site-packages/music21/harmony.py>